

Order Management System

Project Title: Helios Watch Store E-Commerce Database

Skill Task: DBMS

Week: 4

1. Database Design

1.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Order_Date	DATE	NOT NULL
Total	NUMBER(10,2)	NOT NULL
Order_Status	VARCHAR2(20)	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL

1.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	NUMBER	Primary Key
Order_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Quantity	NUMBER	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

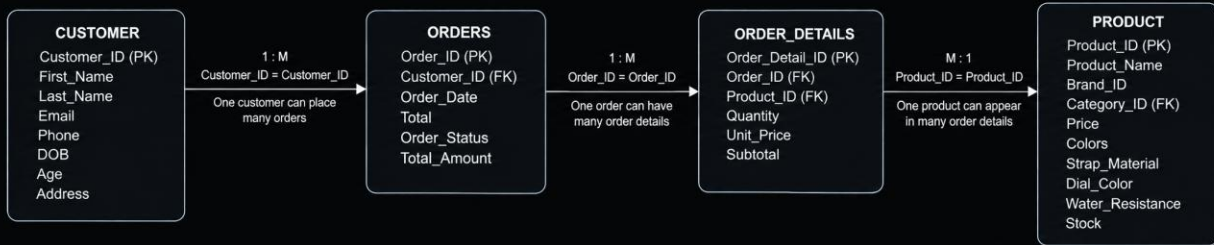
2. Table Relationships

Customer → Orders: One customer can place many orders.

Orders → Order_Details: One order can contain many order details.

Product → Order_Details: One product can appear in many order details.

2. Table Relationships



3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
2     Order_ID      NUMBER PRIMARY KEY,  
3     Customer_ID   NUMBER NOT NULL,  
4     Order_Date    DATE NOT NULL,  
5     Total         NUMBER(10,2) NOT NULL,  
6     Order_Status  VARCHAR2(20) NOT NULL,  
7     Total_Amount  NUMBER(10,2) NOT NULL,  
8     CONSTRAINT FK_Orders_Customer  
9         FOREIGN KEY (Customer_ID)  
10        REFERENCES Customer(Customer_ID)  
11 );
```

Table created.

3.2 Create Order_Details Table

```
SQL> CREATE TABLE Order_Details (  
  2     Order_Detail_ID NUMBER PRIMARY KEY,  
  3     Order_ID         NUMBER NOT NULL,  
  4     Product_ID       NUMBER NOT NULL,  
  5     Quantity         NUMBER NOT NULL,  
  6     Unit_Price       NUMBER(10,2) NOT NULL,  
  7     Subtotal         NUMBER(10,2) NOT NULL,  
  8     CONSTRAINT FK_OrderDetails_Order  
  9         FOREIGN KEY (Order_ID)  
10         REFERENCES Orders(Order_ID),  
11     CONSTRAINT FK_OrderDetails_Product  
12         FOREIGN KEY (Product_ID)  
13         REFERENCES Product(Product_ID)  
14 );
```

Table created.

4. Order Data Insertion

```
SQL> INSERT INTO Orders
  2 VALUES (1001, 1, TO_DATE('01-09-2026','DD-MM-YYYY'), 6497.00, 'Delivered', 6497.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1002, 2, TO_DATE('02-09-2026','DD-MM-YYYY'), 3999.00, 'Shipped', 3999.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1003, 3, TO_DATE('03-09-2026','DD-MM-YYYY'), 1999.00, 'Processing', 1999.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1004, 4, TO_DATE('04-09-2026','DD-MM-YYYY'), 5999.00, 'Delivered', 5999.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1005, 5, TO_DATE('05-09-2026','DD-MM-YYYY'), 7499.00, 'Shipped', 7499.00);

1 row created.
```

Activate Wi

```
SQL>
SQL> INSERT INTO Orders
  2 VALUES (1006, 6, TO_DATE('06-09-2026','DD-MM-YYYY'), 4599.00, 'Processing', 4599.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1007, 7, TO_DATE('07-09-2026','DD-MM-YYYY'), 5299.00, 'Delivered', 5299.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1008, 8, TO_DATE('08-09-2026','DD-MM-YYYY'), 12999.00, 'Shipped', 12999.00);

1 row created.

SQL>
SQL> INSERT INTO Orders
  2 VALUES (1009, 9, TO_DATE('09-09-2026','DD-MM-YYYY'), 2499.00, 'Processing', 2499.00);

1 row created.
```

5. Order_Details Data Insertion

```
SQL> INSERT INTO Order_Details
  2 VALUES (1, 1001, 101, 1, 2499.00, 2499.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (2, 1001, 103, 2, 1999.00, 3998.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (3, 1002, 102, 1, 3999.00, 3999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (4, 1003, 103, 1, 1999.00, 1999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (5, 1004, 104, 1, 5999.00, 5999.00);

1 row created.
```

```
SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (6, 1005, 105, 1, 7499.00, 7499.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (7, 1006, 106, 1, 4599.00, 4599.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (8, 1007, 107, 1, 5299.00, 5299.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (9, 1008, 108, 1, 12999.00, 12999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Details
  2 VALUES (10, 1009, 101, 1, 2499.00, 2499.00);

1 row created.
```

6. Order Modification Operations

6.1 Modify Order Amount

```
SQL> UPDATE Orders
  2  SET Total = 6999.00,
  3      Total_Amount = 6999.00
  4  WHERE Order_ID = 1001;

1 row updated.
```

```
SQL> SELECT *
  2  FROM Orders
  3  WHERE Order_ID = 1001;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL	ORDER_STATUS	TOTAL_AMOUNT
1001	1	01-SEP-26	6999	Delivered	6999

6.2 Modify Order Date

```
SQL> UPDATE Orders
  2  SET Order_Date = TO_DATE('15-09-2026', 'DD-MM-YYYY')
  3  WHERE Order_ID = 1001;

1 row updated.
```

7. Order_Details Modification

```

SQL> UPDATE Order_Details
  2 SET Quantity = 3,
  3     Subtotal = 5997.00
  4 WHERE Order_Detail_ID = 2;

1 row updated.

```

5 rows selected.

```

SQL> SELECT * FROM Order_Details
  2 WHERE Order_Detail_ID = 2;

```

ORDER_DETAIL_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	SUBTOTAL
2	1001	103	3	1999	5997

8. Customer Order History Report

This report displays the customer's details along with their order information.

```

SQL> SELECT
  2     c.Customer_ID,
  3     c.First_Name,
  4     c.Last_Name,
  5     o.Order_ID,
  6     o.Order_Date,
  7     o.Total,
  8     o.Order_Status,
  9     o.Total_Amount
 10 FROM Customer c
 11 JOIN Orders o
 12     ON c.Customer_ID = o.Customer_ID
 13 ORDER BY c.Customer_ID;

```

CUSTOMER_ID	FIRST_NAME	LAST_NAME	ORDER_ID	ORDER_DATE	TOTAL	ORDER_STATUS	TOTAL_AMOUNT
1	Renu	Srinivasan	1001	15-SEP-26	6999	Delivered	6999
2	Arjun	Kumar	1002	02-SEP-26	3999	Shipped	3999

CUSTOMER_ID	FIRST_NAME			

LAST_NAME		ORDER_ID	ORDER_DAT	

TOTAL	ORDER_STATUS	TOTAL_AMOUNT		

3 Priya				
Kumar		1003	03-SEP-26	
1999	Processing	1999		
4 Rahul				
Sharma		1004	04-SEP-26	

CUSTOMER_ID	FIRST_NAME			

LAST_NAME		ORDER_ID	ORDER_DAT	

TOTAL	ORDER_STATUS	TOTAL_AMOUNT		

5999	Delivered	5999		
5 Keerthana				
Mohan		1005	05-SEP-26	
7499	Shipped	7499		
6 Nandhini				

CUSTOMER_ID	FIRST_NAME			

LAST_NAME		ORDER_ID	ORDER_DAT	

TOTAL	ORDER_STATUS	TOTAL_AMOUNT		

Ravi		1006	06-SEP-26	
4599	Processing	4599		
7 Karthik				
Prakash		1007	07-SEP-26	
5299	Delivered	5299		

CUSTOMER_ID	FIRST_NAME			

LAST_NAME		ORDER_ID	ORDER_DAT	

TOTAL	ORDER_STATUS	TOTAL_AMOUNT		

8 Swetha				
Balaji		1008	08-SEP-26	
12999	Shipped	12999		
9 Vignesh				
Ganesh		1009	09-SEP-26	
2499	Processing	2499		

```

CUSTOMER_ID FIRST_NAME
-----
LAST_NAME                                ORDER_ID ORDER_DAT
-----
TOTAL ORDER_STATUS      TOTAL_AMOUNT
-----

9 rows selected.

```

9. Customer-wise Order Summary

This report shows the number of orders and the total amount spent by each customer.

```

SQL> SELECT
  2     c.Customer_ID,
  3     c.First_Name,
  4     c.Last_Name,
  5     COUNT(o.Order_ID) AS Total_Orders,
  6     SUM(o.Total_Amount) AS Total_Amount
  7 FROM Customer c
  8 LEFT JOIN Orders o
  9     ON c.Customer_ID = o.Customer_ID
 10 GROUP BY
 11     c.Customer_ID,
 12     c.First_Name,
 13     c.Last_Name
 14 ORDER BY c.Customer_ID;

```

CUSTOMER_ID	FIRST_NAME		TOTAL_ORDERS	TOTAL_AMOUNT
1	Renu			
	Srinivasan		1	6999
2	Arjun			
	Kumar		1	3999
3	Priya			
	Kumar		1	1999

```
CUSTOMER_ID FIRST_NAME
```

```
-----  
LAST_NAME
```

```
TOTAL_ORDERS TOTAL_AMOUNT  
-----
```

```
4 Rahul  
Sharma 1 5999
```

```
5 Keerthana  
Mohan 1 7499
```

```
6 Nandhini  
Ravi 1 4599
```

```
CUSTOMER_ID FIRST_NAME
```

```
-----  
LAST_NAME
```

```
TOTAL_ORDERS TOTAL_AMOUNT  
-----
```

```
7 Karthik  
Prakash 1 5299
```

```
8 Swetha  
Balaji 1 12999
```

```
9 Vignesh  
Ganesh 1 2499
```

```
9 rows selected.
```